Wearable Smart Emotional Support System Using Electrodermal Activity

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***Abstract*—Stress has become one of the most common problems in modern society and can significantly affect both mental and physical health. Continuous exposure to stress may lead to serious conditions such as anxiety, cardiovascular diseases, and reduced productivity. In many situations, people are unable to recognize or manage their stress levels in real time. With the rapid development of wearable technology and biosensors, physiological signals can now be used to continuously monitor emotional states.**

This paper proposes a wearable smart emotional support system that detects stress using Electrodermal Activity (EDA), also known as Galvanic Skin Response (GSR). The system uses an EDA sensor connected to an ESP32 microcontroller to monitor changes in skin conductance, which are associated with emotional stress. When the system detects increased stress levels, it provides immediate feedback to the user through vibration alerts and breathing guidance displayed on an OLED screen to help the user relax.

In addition, the system includes a safety feature that can share the user's real-time location with predefined contacts during high stress situations. The location is sent as a Google Maps link so that the receiver can easily identify the user's location and provide assistance if required. The proposed system aims to provide a low-cost, portable, and real-time solution for monitoring stress and supporting emotional well-being.

***Index Terms*—Electrodermal Activity (EDA), Stress Detection, Wearable Devices, ESP32, Physiological Monitoring, Internet of Things (IoT), Emotional Health Monitoring**

I. Introduction

Stress is a common experience in everyday life and occurs when individuals face challenging or demanding situations. While short periods of stress can sometimes improve focus and performance, prolonged stress can have negative effects on both mental and physical health. Many people today experience stress due to academic pressure, work responsibilities, and personal challenges. If stress is not recognized and managed early, it may lead to problems such as anxiety, fatigue, reduced productivity, and other health issues.

Traditionally, stress has been assessed using psychological questionnaires or clinical observation. Although these methods can provide useful information, they rely heavily on subjective responses and cannot monitor stress continuously. With the advancement of wearable technologies and biosensors, it has become possible to track physiological signals in real time and gain a better understanding of a person's emotional state.

One of the most widely used physiological signals for stress detection is Electrodermal Activity (EDA), also known as Galvanic Skin Response (GSR). EDA measures changes in the skin's electrical conductance caused by sweat gland activity, which is controlled by the sympathetic nervous system. When a person experiences emotional arousal or stress, the skin conductance level increases, making EDA a useful indicator for detecting stress levels.

Recent developments in wearable devices have made it possible to integrate sensors, microcontrollers, and communication systems into compact and portable devices. These wearable systems can continuously monitor physiological signals and provide real-time feedback to help users become aware of their emotional state and take steps to relax.

In this work, we propose a wearable smart emotional support system that detects stress using electrodermal activity signals. The system uses an ESP32 microcontroller to process signals from an EDA sensor and determine changes in stress levels. When stress is detected, the device alerts the user through a vibration motor and displays breathing guidance on an OLED screen to help the user relax. In addition, the system includes a safety feature that can share the user's location with predefined contacts during high stress situations, allowing quick assistance if required.

II. Related Work

In recent years, researchers have explored the use of wearable devices and physiological signals for stress detection. Electrodermal Activity (EDA) is one of the most widely used signals for monitoring emotional arousal because it directly reflects changes in the sympathetic nervous system. Several studies have demonstrated that wearable sensors can effectively capture EDA signals and use them to analyze stress patterns.

Sano and Picard proposed a wearable sensing system that combines physiological signals and machine learning techniques to recognize stress levels. Their system used EDA and other biosignals collected through wearable sensors to analyze emotional states in real time. Similarly, Gjoreski et al. developed a continuous stress detection system using a wrist-

worn device that integrates multiple sensors such as EDA, motion, and temperature to improve detection accuracy.

Other studies have focused on analyzing stress in real-world environments. Hernandez et al. investigated stress detection using physiological signals collected from wearable devices during daily activities. Their research showed that combining multiple physiological signals can improve the reliability of stress recognition systems.

Although many of these systems provide accurate stress detection, they often require complex hardware setups, high computational resources, or smartphone-based processing. In contrast, the proposed system focuses on a simple and low-cost wearable solution that can detect stress using EDA signals and provide immediate feedback to the user. In addition, the system includes a location alert feature that allows emergency contacts to be notified during high stress situations.

III. Proposed System

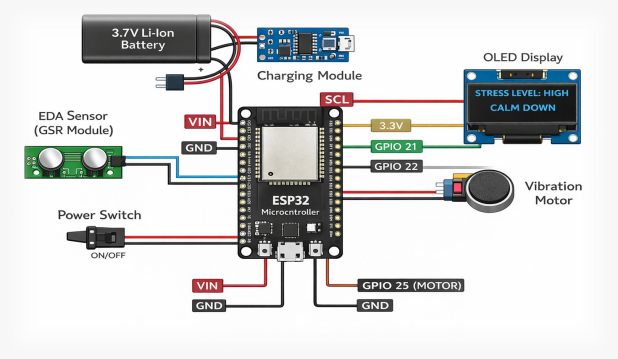


Fig. 1. Hardware architecture of the proposed wearable stress detection system

The proposed system is a wearable smart emotional support device designed to detect stress using Electrodermal Activity (EDA) signals and provide immediate assistance to the user. The system continuously monitors the user's skin conductance using an EDA sensor. Variations in the skin conductance level are associated with emotional changes such as stress, anxiety, or excitement.

The EDA sensor is connected to an ESP32 microcontroller, which acts as the main processing unit of the system. The ESP32 reads the analog signals from the sensor and processes them to determine whether the user is experiencing elevated stress levels. A baseline value is first calculated to represent the user's normal skin conductance level. The system then compares real-time sensor readings with this baseline to identify abnormal changes.

When stress is detected, the system provides immediate feedback to the user through two mechanisms. First, a vibration motor is activated to notify the user that an elevated stress level has been detected. Second, the OLED display shows breathing guidance instructions that encourage the user to follow a controlled breathing pattern in order to relax.

In addition to providing emotional support, the system includes a safety feature that can send the user's location to predefined contacts. When high stress levels persist for a certain duration, the system triggers a notification that shares the user's location as a Google Maps link. This feature allows friends or family members to identify the user's location and provide assistance if necessary.

The proposed design focuses on developing a simple, low-cost, and portable wearable system that can continuously monitor emotional stress and provide both immediate feedback and emergency support.

IV. Methodology

The methodology of the proposed system focuses on continuously monitoring electrodermal activity signals and analyzing them to detect stress levels. The system operates through several stages including signal acquisition, signal processing, stress detection, and user feedback. The overall workflow of the proposed stress detection system is illustrated in Fig. 2.

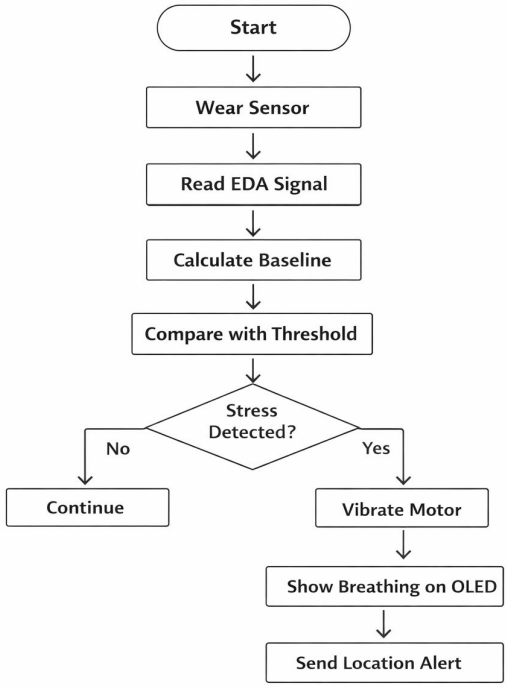


Fig. 2. Workflow of stress detection and location alert mechanism

A. Signal Acquisition

Electrodermal Activity (EDA) signals are captured using a GSR sensor placed on the user's skin. The sensor measures changes in skin conductance caused by sweat gland activity. These variations occur due to changes in the sympathetic

nervous system during emotional responses such as stress or anxiety. The analog signal generated by the sensor is transmitted to the ESP32 microcontroller for further processing.

B. Signal Processing

The ESP32 microcontroller reads the analog signal from the EDA sensor using its built-in analog-to-digital converter (ADC). Since raw physiological signals may contain noise and fluctuations, basic filtering techniques are applied to smooth the signal. The system calculates an average value from multiple sensor readings to obtain a more stable representation of the user's electrodermal activity.

C. Stress Detection

To identify stress levels, the system first determines a baseline value representing the user's normal skin conductance level. This baseline is calculated during an initial calibration phase when the user is in a relaxed state. Since electrodermal activity varies between individuals, the baseline value is personalized for each user. This individualized calibration allows the system to accurately detect abnormal increases in skin conductance that may indicate emotional stress.

During operation, the system continuously compares real-time EDA values with the baseline. If the difference between the current value and the baseline exceeds a predefined threshold for a certain duration, the system identifies the condition as elevated stress.

D. Feedback Mechanism

When stress is detected, the system provides immediate feedback to help the user relax. A vibration motor is activated to notify the user that increased stress levels have been detected. At the same time, the OLED display shows breathing guidance instructions such as inhale and exhale prompts to encourage controlled breathing. This biofeedback mechanism helps the user regulate their breathing and reduce stress.

E. Location Alert System

In addition to providing immediate feedback, the system also includes a safety feature for emergency situations. If high stress levels persist for an extended period, the system triggers a location alert mechanism. The user's location is obtained through the connected mobile device using GPS services and automatically converted into a Google Maps link.

An automation workflow integrated with a Telegram bot is used to transmit the alert. When the stress threshold remains exceeded for a specified duration, the automation system is triggered and sends the location message to predefined emergency contacts. This message allows the receiver to directly open the location in Google Maps and quickly identify the user's position in order to provide assistance if necessary.

V. Hardware and Software Implementation

The proposed wearable emotional support system integrates both hardware and software components to detect stress and provide feedback to the user.

A. Hardware Implementation

The hardware prototype consists of several key components including an ESP32 microcontroller, an Electrodermal Activity (EDA) sensor, an OLED display, and a vibration motor.

The EDA sensor measures variations in skin conductance from the user's fingers. These variations occur due to sweat gland activity, which is controlled by the sympathetic nervous system. The sensor outputs analog signals corresponding to the user's electrodermal activity.

The ESP32 microcontroller acts as the main processing unit of the system. It reads the analog signals from the EDA sensor using its built-in analog-to-digital converter (ADC) and processes the data to determine stress levels.

An OLED display is used to provide visual feedback to the user. When stress is detected, the display shows breathing guidance instructions such as "Breathe In" and "Breathe Out" to help the user regulate breathing and relax.

A vibration motor is also connected to the ESP32 to provide haptic feedback. When the stress level exceeds a predefined threshold, the motor vibrates to alert the user.

B. Software Implementation

The system software is developed using the Arduino IDE. The program continuously reads electrodermal activity signals from the EDA sensor and processes them to determine stress levels.

Initially, the system performs a calibration phase to calculate a baseline skin conductance value for the user while the user remains in a relaxed state. Since electrodermal activity varies between individuals, this baseline value is personalized for each user. During normal operation, real-time EDA readings are continuously compared with this individualized baseline value. If the difference between the current reading and the baseline exceeds a predefined threshold for several consecutive readings, the system identifies the condition as elevated stress.

Once stress is detected, the ESP32 activates the vibration motor and displays breathing guidance on the OLED screen to encourage controlled breathing. If high stress levels persist for a longer duration, an automated alert mechanism is triggered.

The system integrates a Telegram bot with a mobile automation workflow to transmit emergency alerts. When the stress threshold remains exceeded for a specified duration, the automation process retrieves the user's current GPS coordinates from the connected mobile device. These coordinates are automatically converted into a Google Maps link and sent through the Telegram bot to predefined emergency contacts. This enables the receiver to immediately view the user's location and provide assistance if necessary.

VI. Experimental Setup

To evaluate the performance of the proposed wearable emotional support system, a prototype device was developed and tested under different conditions.

The prototype consisted of an ESP32 microcontroller, an Electrodermal Activity (EDA) sensor, an OLED display, and



Fig. 3. Prototype wearable stress detection device developed for experimentation

a vibration motor. The EDA sensor electrodes were placed on the user's fingers to measure variations in skin conductance.

During the experiment, the system first entered a calibration phase where the baseline electrodermal activity of the user was measured while the user remained in a relaxed state. This baseline value represented the normal skin conductance level for the user.

After calibration, the system continuously monitored the EDA signal in real time. Different emotional conditions were simulated during testing. For example, the user was asked to perform tasks that could induce mild stress such as solving time-restricted problems or performing mental calculations.

The system continuously compared real-time EDA values with the baseline value. If the difference exceeded the predefined threshold for several consecutive readings, the system classified the condition as elevated stress.

Once stress was detected, the device activated the vibration motor and displayed breathing guidance on the OLED display. If the elevated stress condition persisted for a longer duration, the system triggered the emergency alert feature.

The emergency alert system sent a notification message using a Telegram bot. The message included a Google Maps location link, allowing predefined contacts to easily identify the user's location and provide assistance if necessary.

VII. Results and Discussion

The proposed wearable emotional support system was tested to evaluate its ability to detect stress and provide timely feedback to the user. The prototype device continuously monitored electrodermal activity signals and analyzed changes in skin conductance associated with emotional stress.

During testing, the system first performed a calibration stage to determine the baseline electrodermal activity of the user while the user remained in a relaxed state. This baseline value represented the normal skin conductance level for the user. After calibration, the device continuously monitored real-time EDA signals and compared them with the baseline value.

When the difference between the real-time EDA value and the baseline exceeded the predefined threshold for several consecutive readings, the system identified the condition as elevated stress. Once stress was detected, the device activated the vibration motor to notify the user. At the same time, the OLED display provided breathing guidance instructions that encouraged the user to follow a controlled breathing pattern in order to relax. Fig. 4 shows the breathing guidance displayed on the OLED screen.

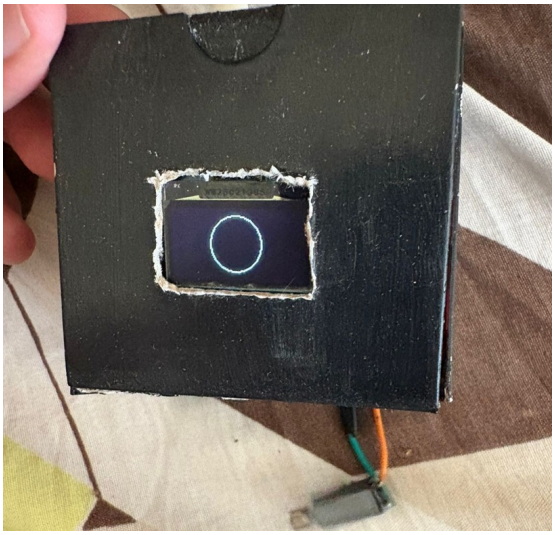


Fig. 4. OLED display showing breathing guidance during stress detection

Experimental observations indicated that electrodermal activity values increased noticeably during stressful conditions compared to relaxed states. The system was able to detect these variations and respond quickly by triggering the feedback mechanism. The breathing guidance displayed on the OLED screen helped the user regulate breathing and gradually return to a calmer state.

The emergency alert functionality of the system was also tested. When elevated stress levels persisted for an extended duration, the system triggered a notification message through a Telegram bot. Fig. 5 shows the stress alert notification generated by the system.

The alert message contained a Google Maps link representing the user's location. This location alert allows predefined contacts to easily identify the user's position and provide assistance if required, as shown in Fig. 6.

Overall, the prototype demonstrated reliable performance in detecting stress and providing both emotional support and safety notifications. The results suggest that electrodermal activity is a useful physiological signal for monitoring emotional stress in wearable devices.

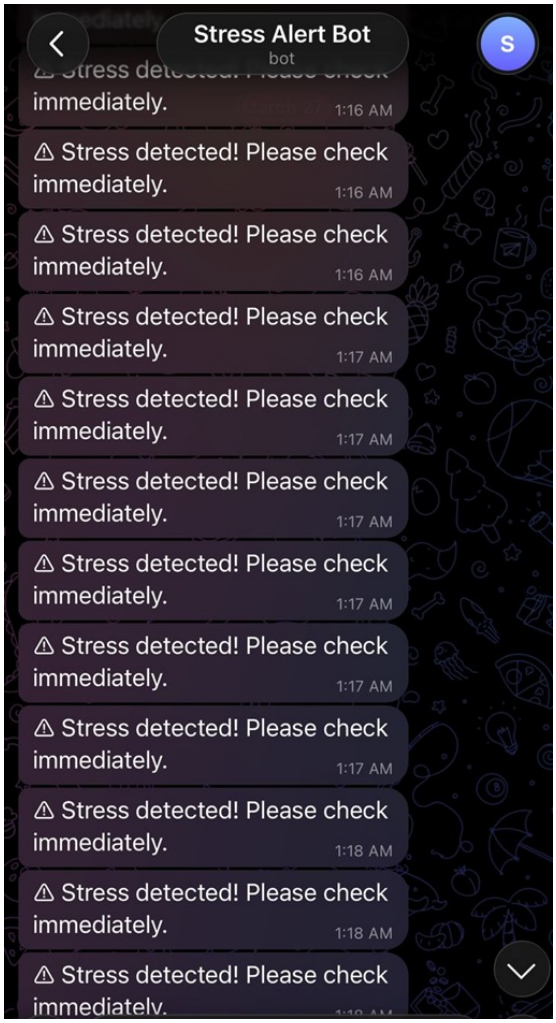


Fig. 5. Stress alert notification generated by the system

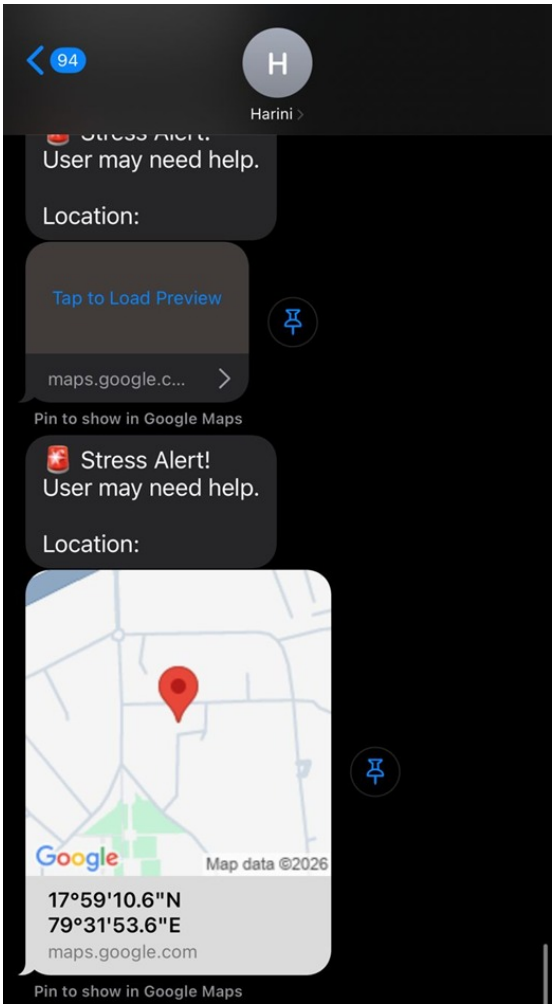


Fig. 6. Location alert message containing Google Maps link sent to emergency contact

VIII. Conclusion

In this paper, a wearable smart emotional support system based on Electrodermal Activity (EDA) was proposed and implemented to detect stress levels in real time. The system uses an EDA sensor to measure variations in skin conductance associated with emotional stress. These signals are processed using an ESP32 microcontroller to identify abnormal changes in the user's stress level.

When elevated stress levels are detected, the device provides immediate feedback through vibration alerts and breathing guidance displayed on an OLED screen. This feedback helps the user become aware of their emotional state and encourages controlled breathing to reduce stress. In addition to emotional support, the system also incorporates an automated location alert mechanism that shares the user's GPS coordinates with predefined contacts through a Telegram-based notification system when high stress levels persist.

The prototype implementation demonstrates that electrodermal activity can be effectively used as a physiological indicator for monitoring emotional stress. The proposed system offers a low-cost, portable, and real-time solution for stress monitoring and personal safety.

Future work may focus on improving stress detection accuracy by incorporating additional physiological signals such as heart rate or skin temperature. Machine learning techniques could also be integrated to enhance the system's ability to classify different emotional states more accurately.

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